

THE IDEA

- Replace MLD's text conditioning with the vehicle ego trajectory.
- Feed the denoiser ALL 196 per-timestep ego tokens — not one pooled summary — so every denoising step has temporal resolution over the trajectory.
- Ego encoder is contrastively pretrained, then FROZEN; only the denoiser trains.

MAIN RESULTS · UNIFIED EVALUATION, 3 REPLICATIONS

System	FID ↓	R@1	MM	Div
EgoPed-IA (+ interaction sampling)	2.880	0.323	3.37	5.67
EgoPed (full-sequence conditioning)	3.392	0.548	3.96	5.79
Unfrozen encoder	5.079	0.352	2.62	6.06
Pooled-token baseline	5.620	0.665	3.71	5.82
Larger latent (8 vs 4)	6.677	0.663	3.37	5.89
Ground truth	—	—	—	5.50

ARCHITECTURE VS. DATA CURATION · 2×2

FID ↓	– interaction sampling	+ interaction sampling
Pooled	6.214	5.620
Full-sequence	3.392	2.880

Conditioning granularity dominates (≈2.7 FID in both columns, ~5× the data effect); the two effects are additive → the headline is conservative, not inflated.

IN FLIGHT: information ladder — 0 bits (trained unconditional: FID 3.24) → 5.17 bits vehicle-only text (training) → 6.09 bits oracle text incl. GT behaviour (2.82) → continuous trajectory (H4 3.39 / IA 2.88). FID ranks marginals, not conditioning — see ADE/FDE + behaviour. **Open:** pose-label quality audit · EgoPed-IA is single-seed · no physics constraints · no perceptual study yet.

WHAT ACTUALLY MATTERS

Question	Finding	Verdict
Unfreeze the ego encoder?	MM 3.96→2.62	freezing critical
Skip encoder pretraining?	R@1 = chance	≈ unconditional
Cross- instead of self-attention?	3.49 vs 3.34	no difference (p=.33)
Bigger VAE latent (8 vs 4)?	6.68 vs 5.62	19% worse
Different training seed?	3.39/4.72/4.80	real variance

EXTERNAL BASELINES · CAPABILITY LADDER

Approach	FID ↓	Reading
EgoPed-IA (ours)	2.88	in-domain, ego-conditioned
Unconditional, properly trained	3.24	≈ H4 — FID is blind to conditioning
Unconditional prior (ego zeroed)	5.18	a weak null branch, not a prior
Pretrained text-to-motion MLD	14.07	T2M does not transfer
Raw-space diffusion (MDM-style)	26.43	latent backbone justified
Ego→motion regressor	35.47	L2-optimal, unrealistic
Trajectory + static body	47.7	needs articulated motion

VALIDATED ON FOUR INDEPENDENT AXES

- FID — 40% over pooled (non-overlapping CIs); but a trained prior also reaches 3.24, so FID alone does not credit conditioning.
- Held-out split — scene-disjoint, never used for selection: ranking preserved.
- Trajectory ADE/FDE — no learned evaluator: beats pooled on all 4 metrics despite lower R@1 → the R@1 gap is an embedding artifact.
- Behavioral probe — best coupling of stop/walk decisions to the ego; unconditional model is ego-blind (sanity ✓).